COMP/EECE 7/8740 Neural Networks

Topics:

- CNN for Segmentation (regression) tasks
  - Transposed convolutions
  - Segmentation models
- Demos of segmentation model

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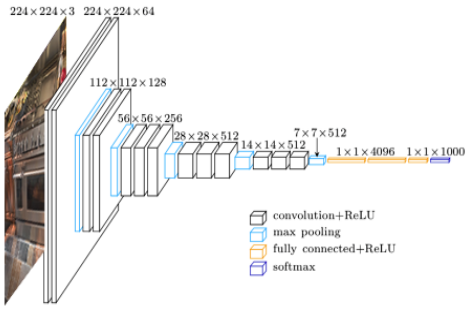
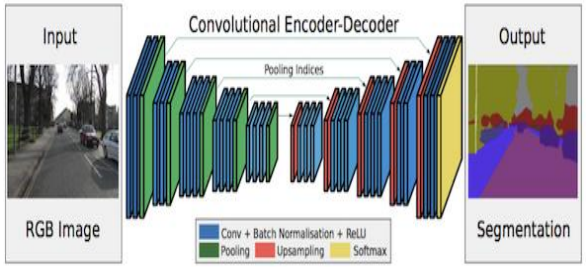
Deep CNN models and applications

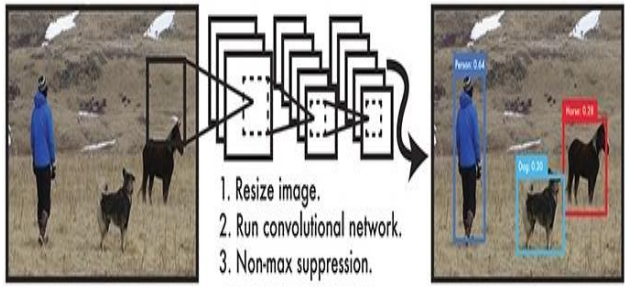
- **Classification**



- **Segmentation(regression) models**



- **Detection models**



Models for classification:

- AlexNet
- VGG Net
- GoogleNet
- ResNet
- Inception-ResNet
- DenseNet
- FractalNet
- CapsuleNet and
- IRRCNN

Models for Segmentation:

- FCN
- SegNet
- Dialted Convolution
- RefineNet
- Pyramid Scene Parsing (PSP):PSPNet
- DeepLab
- U-Net

Models for Detection:

- Region based CNN (RCNN)
- Fast RCNN
- Faster RCNN
- Mask RCNN
- You Only Look Once (YOLO)
- Single Short Multibox Detector (SSD)

Seg.: http://blog.qure.ai/notes/semantic-segmentation-deep-learning-review

Detection: https://towardsdatascience.com/

Regression or Segmentation model?

- Mathematically it performs pixel-wise regression or classification **depending on the output and loss function.**
- This type of models (such as U-Net) are designed to perform **image-to-image mapping**, where:

$$f(x) = g(x)/x$$

  - **Input (x):** Image
  - **Output (y = g(x)):** Segmentation mask (same spatial size as input)

- Instead of predicting **a single value**, it predicts **a label/value for every pixel** and categorized as a **semantic segmentation network**.

Can We Call U-Net a Regression Model?

- **Yes — *in a mathematical sense***
- If:
  - The output mask is **continuous-valued**, and
  - You use **regression losses** like:
    - Mean Squared Error (MSE)
    - Mean Absolute Error (MAE)
    - Smooth L1 loss

- Then U-Net is effectively performing **pixel-wise regression**:

$$g(x) = \hat{y}, \qquad \hat{y} \in \mathbb{R}^{H \times W}$$

Can We Call U-Net type model a Regression Model?

- When you apply this model to solve problem as follows :
  - Depth map estimation
  - Denoising
  - Super-resolution
  - Image reconstruction
  - Heatmap prediction
  - Medical imaging:
    - Predicting **continuous tissue probability maps**

- In these cases, **U-Net is a dense regression model**.

When Is U-Net a Classification Model?

- If:
  - Output mask contains **class labels**
  - Uses **Cross-Entropy / Dice / Focal loss**
  - Output is **probabilities per pixel**

- Then it is performing **pixel-wise classification**, i.e., **segmentation**.

- **Example:**
  - Tumor vs background
  - Multi-class tissue segmentation
  - Organ segmentation

Clear Comparison

Task Type	**Output**	**Loss**	**Model Interpretation**
Semantic Segmentation	Class per pixel	Cross-entropy, Dice	Pixel-wise classification
Image Regression	Continuous value per pixel	MSE, MAE	Dense regression
Reconstruction	Image	L1, L2	Autoencoder-style regression

What is semantic segmentation

- **Semantic Segmentation** : is a technique that detects , for each pixel , the object category it belongs to , all object categories (labels) must be known to the model.

- Pixel-level classification

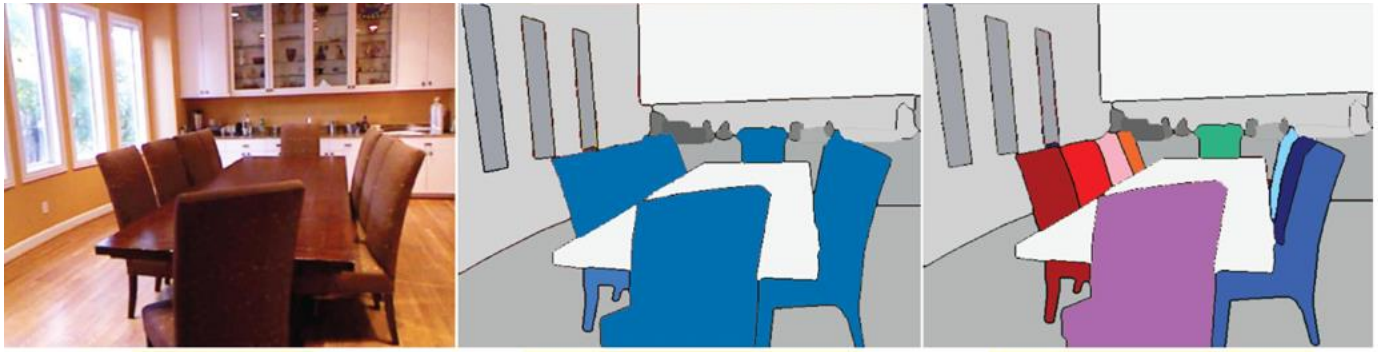
(Badrinarayanan, Kendall, & Cipolla, 2015)

(Chen, Papandreou, Kokkinos, Murphy, & Yuille, 2014)

What is instance segmentation?

- **Instance Segmentation** : same as semantic segmentation, but dives a bit deeper, **it identifies , for each pixel, the object instance it belongs to**.

- **Semantic vs Instance Segmentation** :
  - The main difference is that **differentiates two objects with the same labels** in comparison to semantic segmentation.



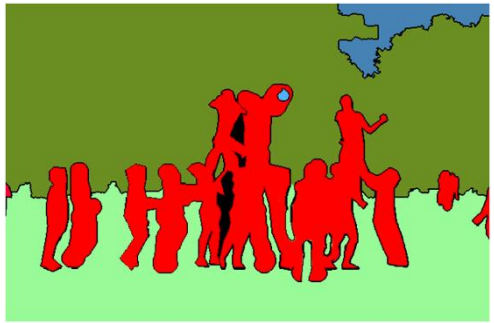
Input Image | Semantic Segmentation | Instance Segmentation

What is panoramic segmentation?

- Combination annotation of **semantic segmentation and instance segmentation, more complex.**

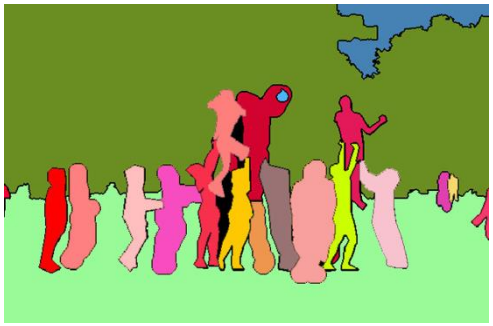
- Panoramic segmentation means that all targets should be detected and **different instances in the same category should be distinguished.**

Input



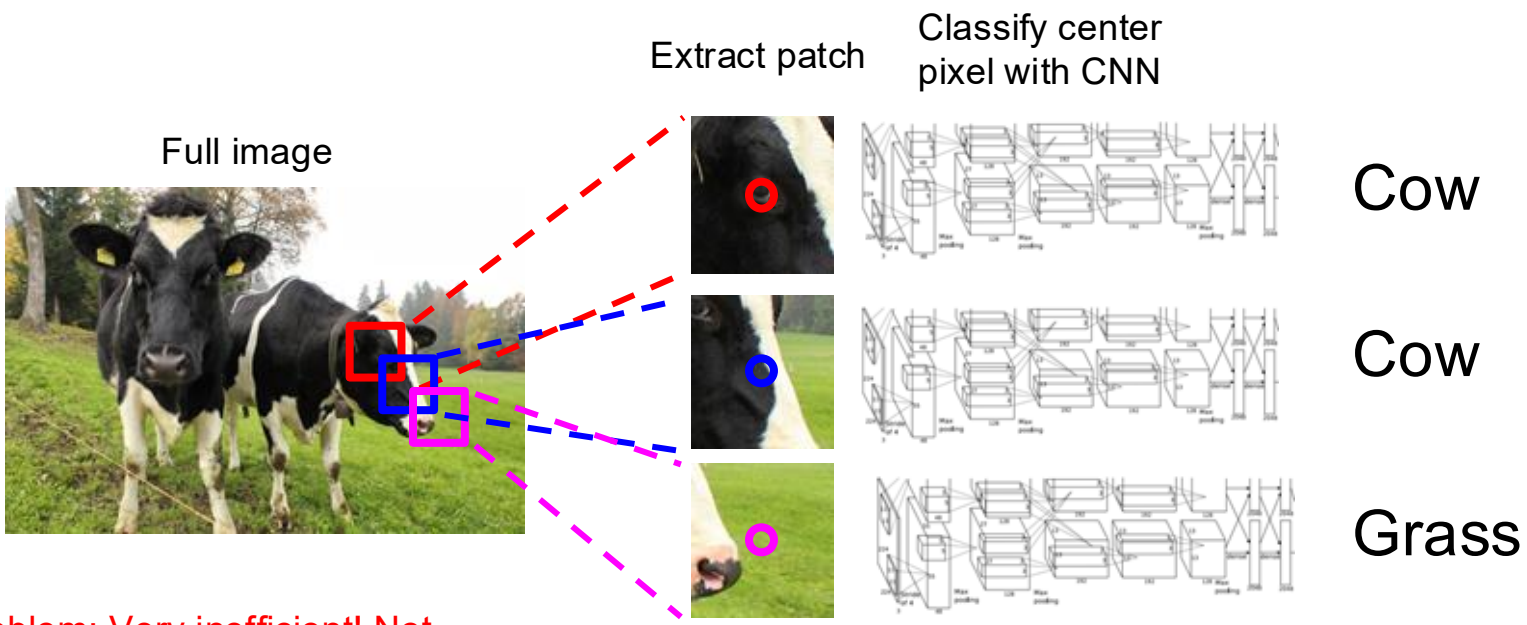
Semantic segmentation

Instance segmentation



Panoramic segmentation

Semantic Segmentation Idea: sliding window

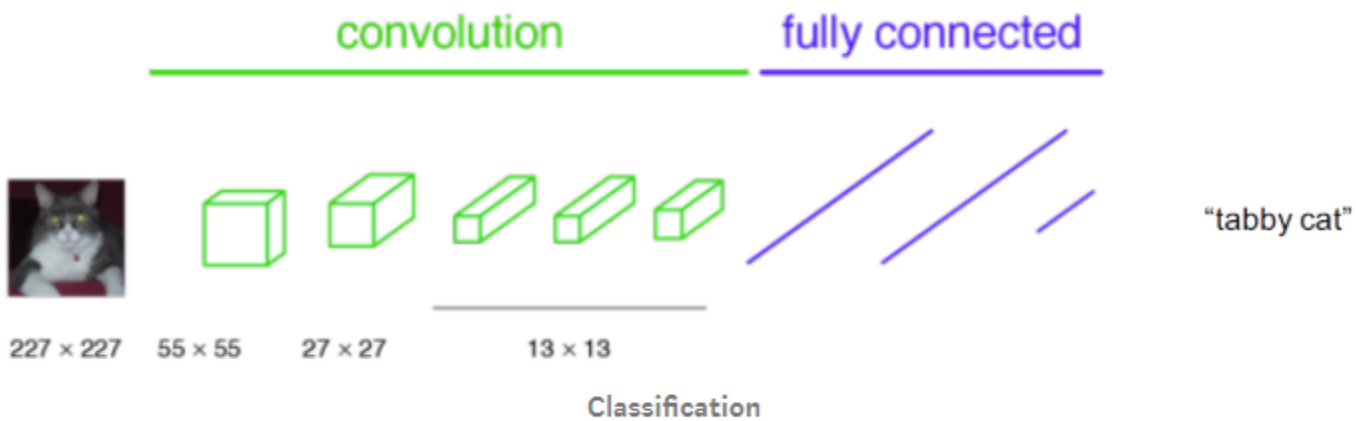


Problem: Very inefficient! Not reusing shared features between overlapping patches

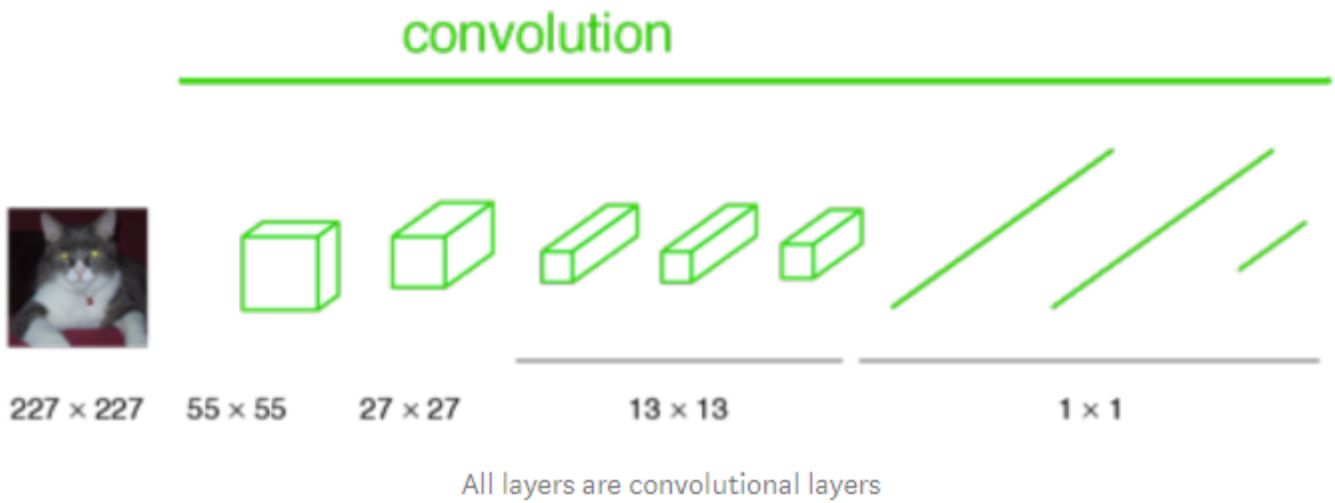
Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

From Image Classification to Semantic Segmentation



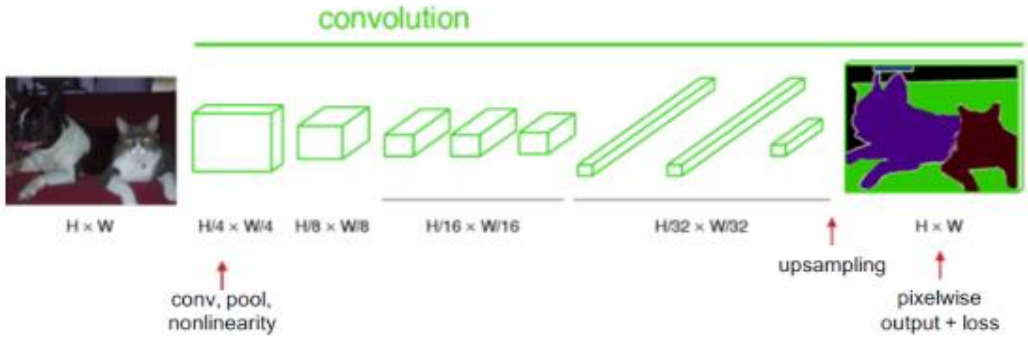
Imagine we turn the FC layers into 1 × 1 convolutional layers:



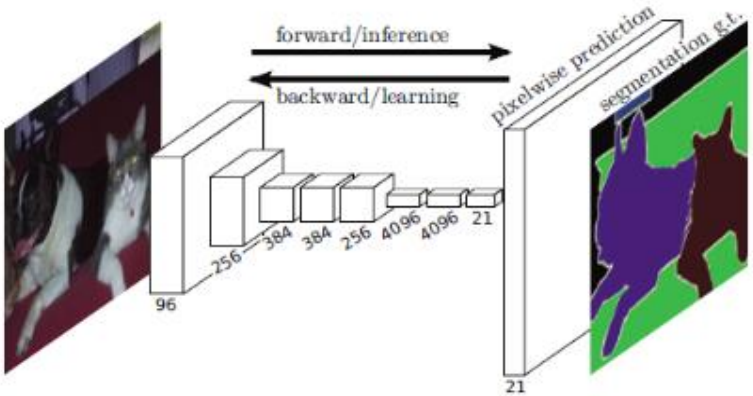
https://towardsdatascience.com/review-fcn-semantic-segmentation-eb8c9b50d2d1

Fully Convolution Network (FCN)

- If the image is not downsized, the output will not be a single label.
- Instead, the output has a size smaller than the input image (due to the max pooling)



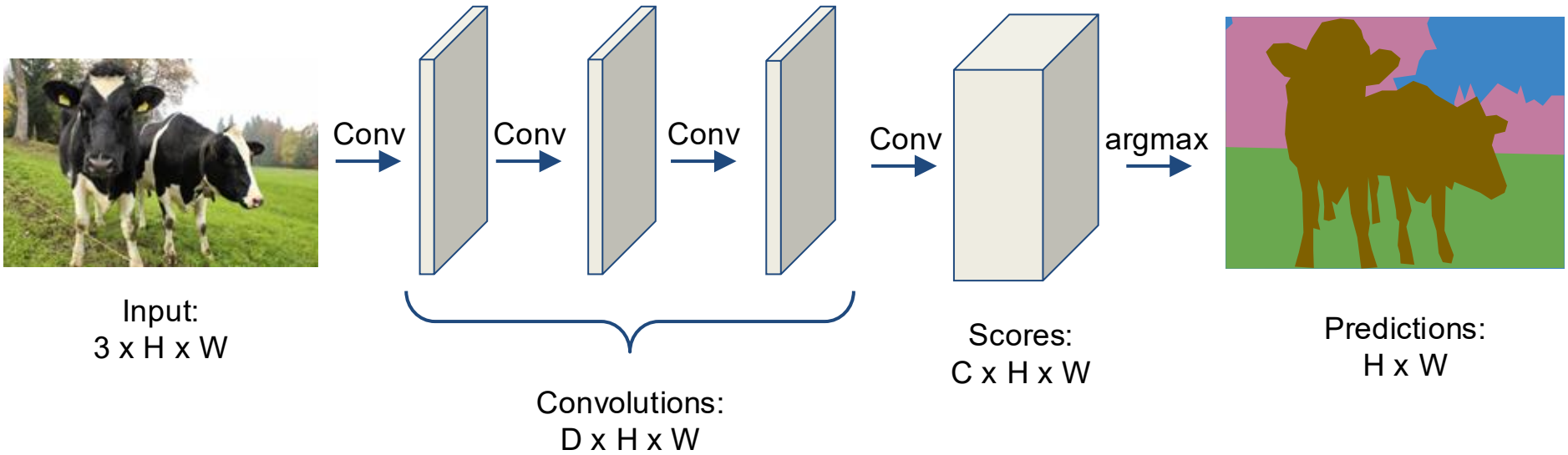
Upsampling at the last step



Feature Map / Filter Number Along Layers

Semantic Segmentation Idea: Fully Convolutional

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Semantic Segmentation Idea: Fully Convolutional

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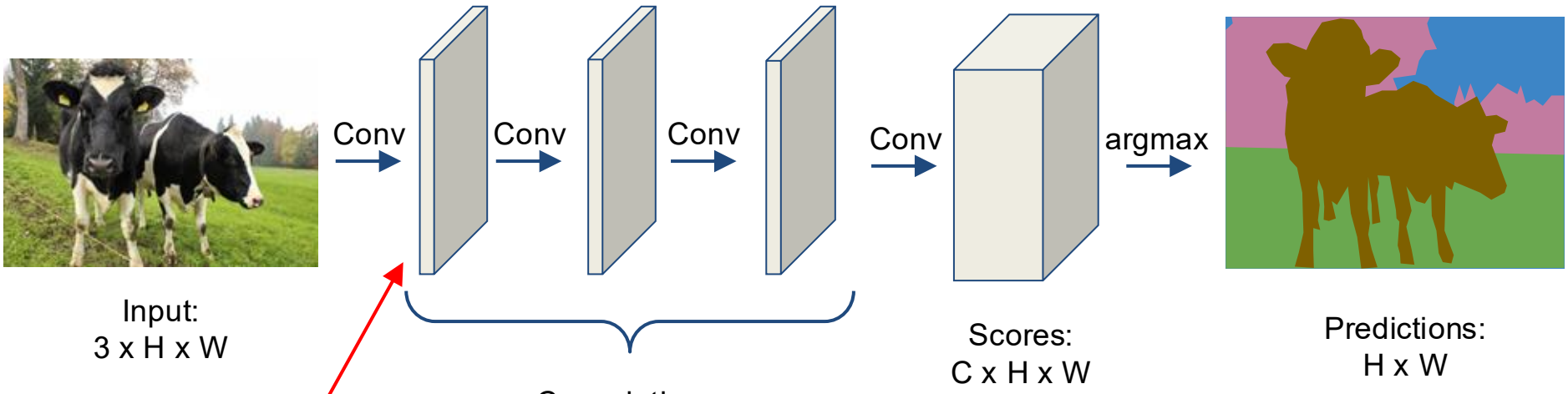
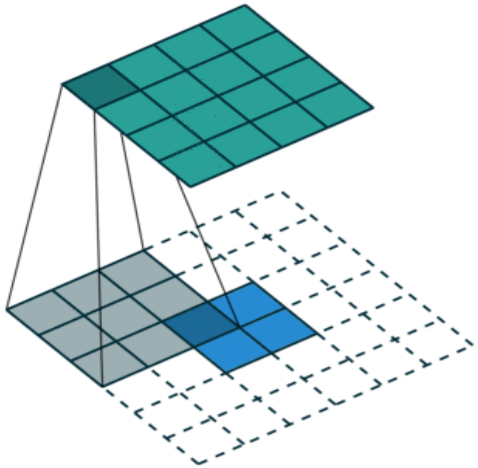


Problem: convolutions at original image resolution will be very expensive ...

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Convolutions for Semantic Segmentation

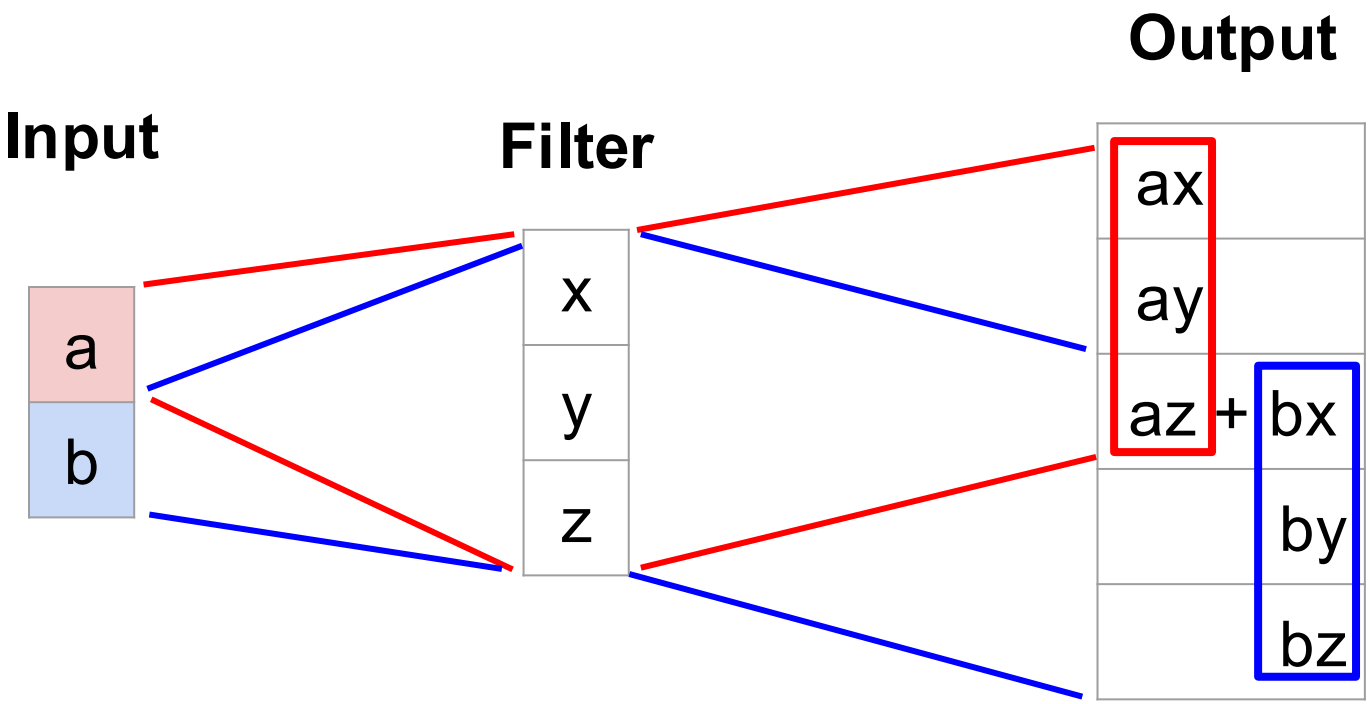
- Up-sampling with Deconvolution
  - Up convolution
  - Transposed convolution
  - Fractional stride convolution
- Up-sampling and fusing the output feature maps.



Upsampling Via Deconvolution (Blue: Input, Green: Output)

- https://towardsdatascience.com/review-fcn-semantic-segmentation-eb8c9b50d2d1
- https://distill.pub/2016/deconv-checkerboard/

Learnable Up-sampling : 1D example



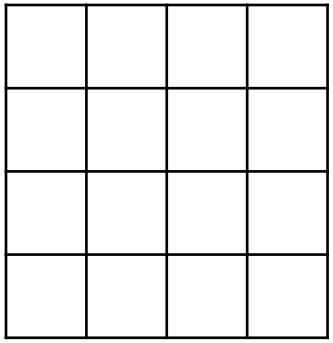
Output contains copies of the **filter weighted by the input**, summing at where at overlaps in the output

Need to crop one pixel from output to make output exactly 2x input

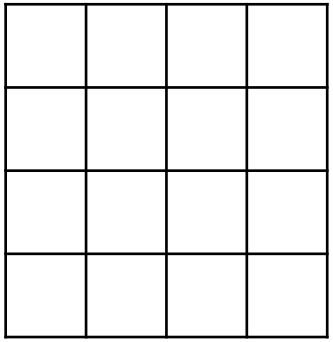
Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Learnable Upsampling: Transpose Convolution

Recall: Typical 3 x 3 convolution, stride 1 pad 1



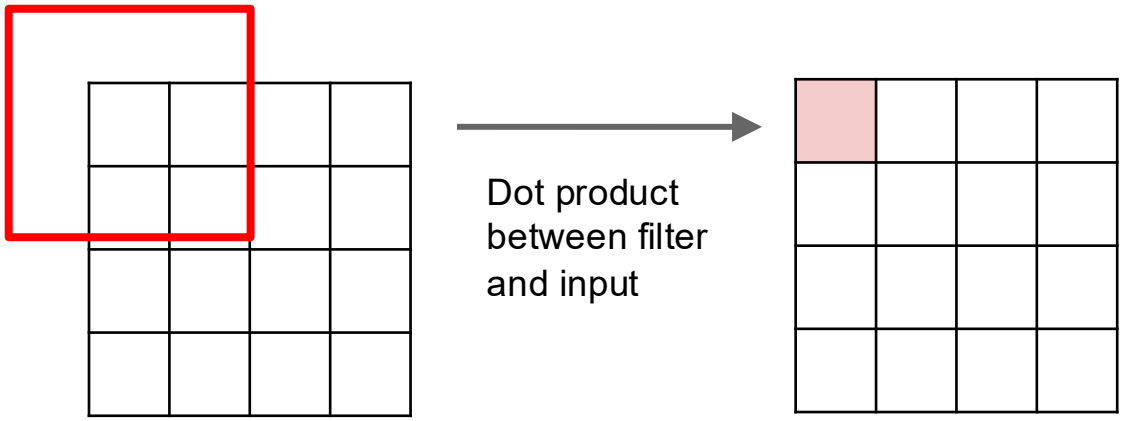
Input: 4 x 4



Output: 4 x 4

Learnable down-sampling : Same Convolution

Recall: Normal 3 x 3 convolution, stride 1 pad 1



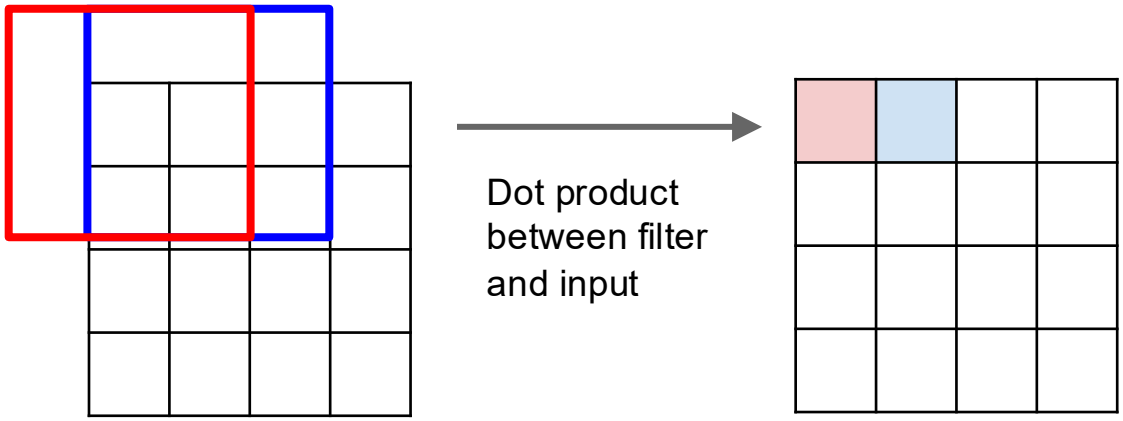
Input: 4 x 4

Output: 4 x 4

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Learnable down-sampling : Same Convolution

Recall: Normal 3 x 3 convolution, stride 1 pad 1

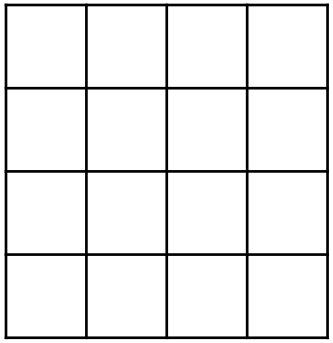


Input: 4 x 4

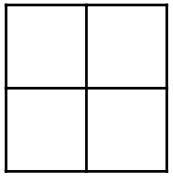
Output: 4 x 4

Learnable down-sampling : Valid Convolution

Recall: Normal 3 x 3 convolution, <u>stride 2</u> pad 1



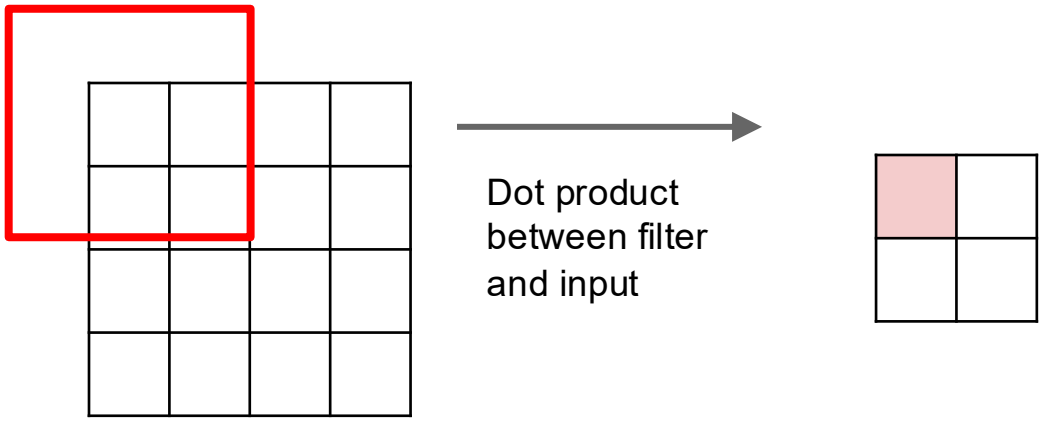
Input: 4 x 4



Output: 2 x 2

Learnable down-sampling: Valid Convolution with stride 2

Recall: Normal 3 x 3 convolution, <u>stride 2</u> pad 1

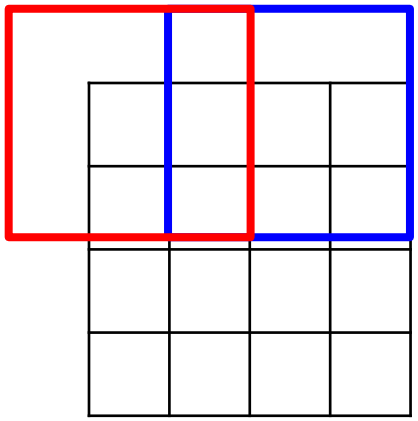


Input: 4 x 4

Output: 2 x 2

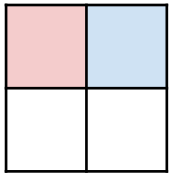
Learnable down-sampling: Valid Convolution with stride 2

Recall: Normal 3 x 3 convolution, <u>stride 2</u> pad 1



Input: 4 x 4

Dot product between filter and input



Output: 2 x 2

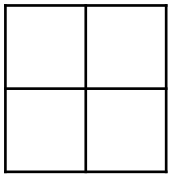
Filter moves 2 pixels in the input for every one pixel in the output

Stride gives ratio between movement in input and output

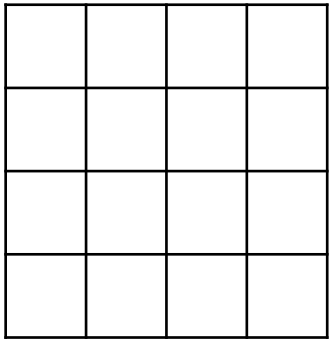
Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Learnable Upsampling: **Transpose Convolution**

3 x 3 **transpose convolution**, stride 2 pad 1



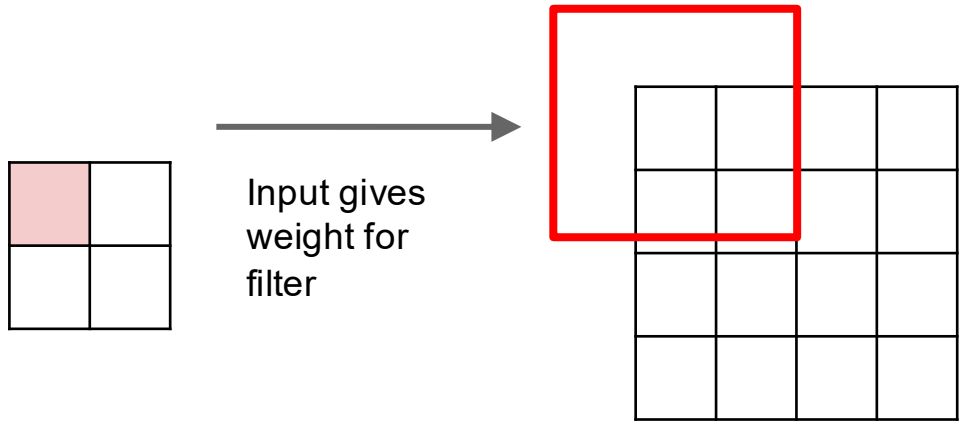
Input: 2 x 2



Output: 4 x 4

Learnable Upsampling: Transpose Convolution

3 x 3 **transpose** convolution, stride 2 pad 1



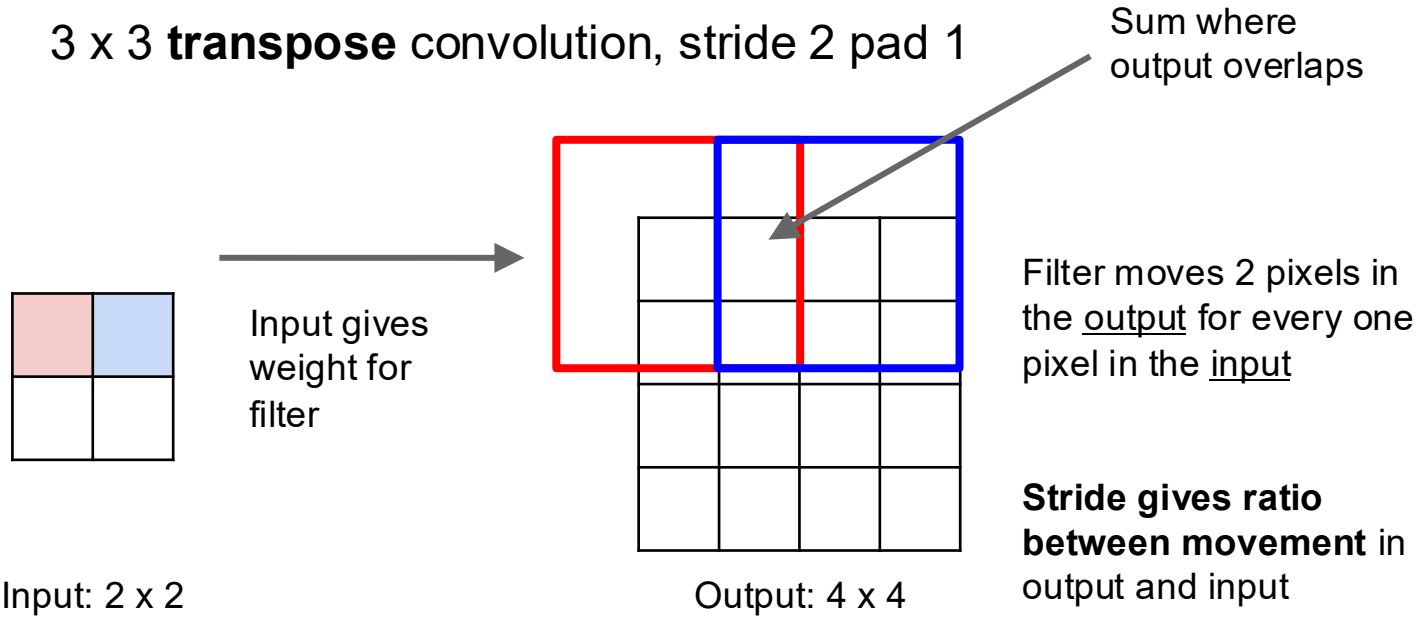
Input: 2 x 2

Output: 4 x 4

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Learnable Upsampling: Transpose Convolution

3 x 3 **transpose** convolution, stride 2 pad 1



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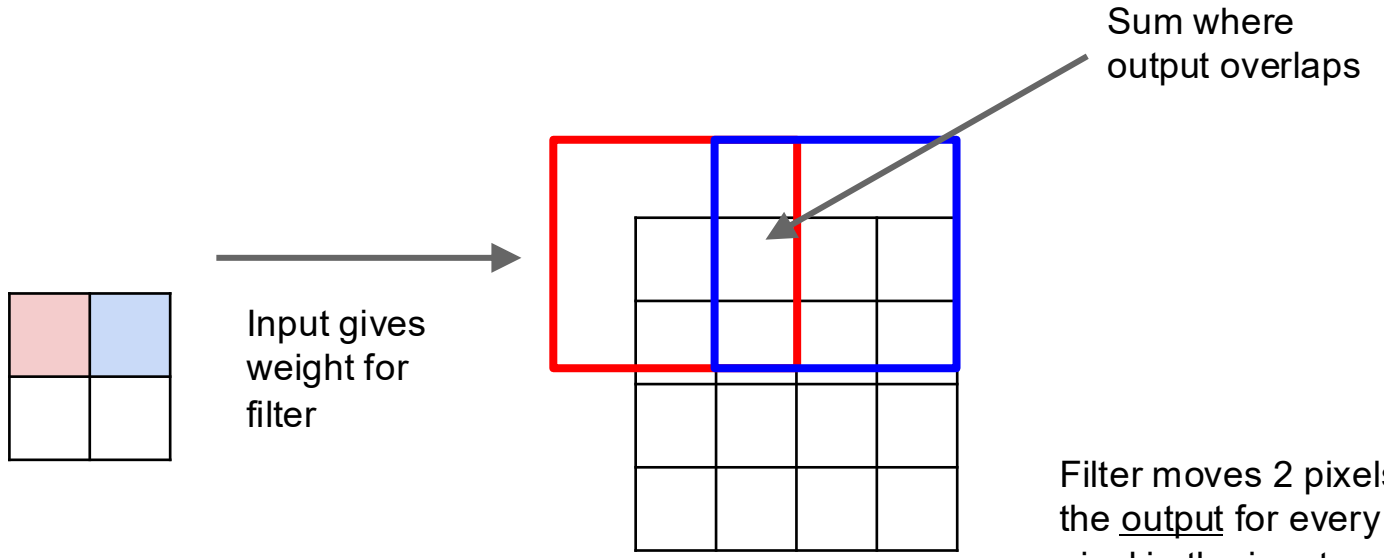


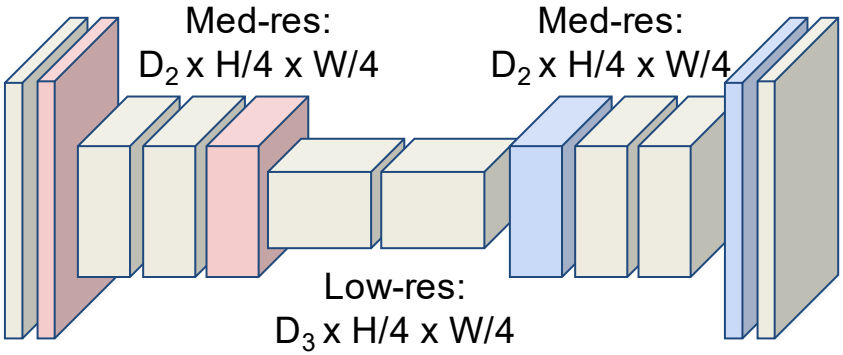
Filter moves 2 pixels in the <u>output</u> for every one pixel in the <u>input</u>

Stride gives ratio between movement in output and input

Semantic Segmentation Idea: Fully Convolutional

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Input:
3 x H x W



High-res:
D_1 x H/2 x W/2

High-res:
D_1 x H/2 x W/2

Predictions:
H x W

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

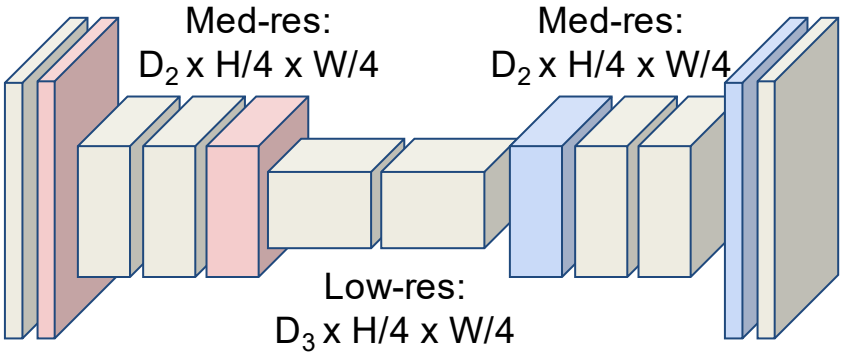
Semantic Segmentation Idea: Fully Convolutional

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Downsampling: Pooling, strided convolution

Upsampling: ???

Input: 3 x H x W



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Predictions: H x W

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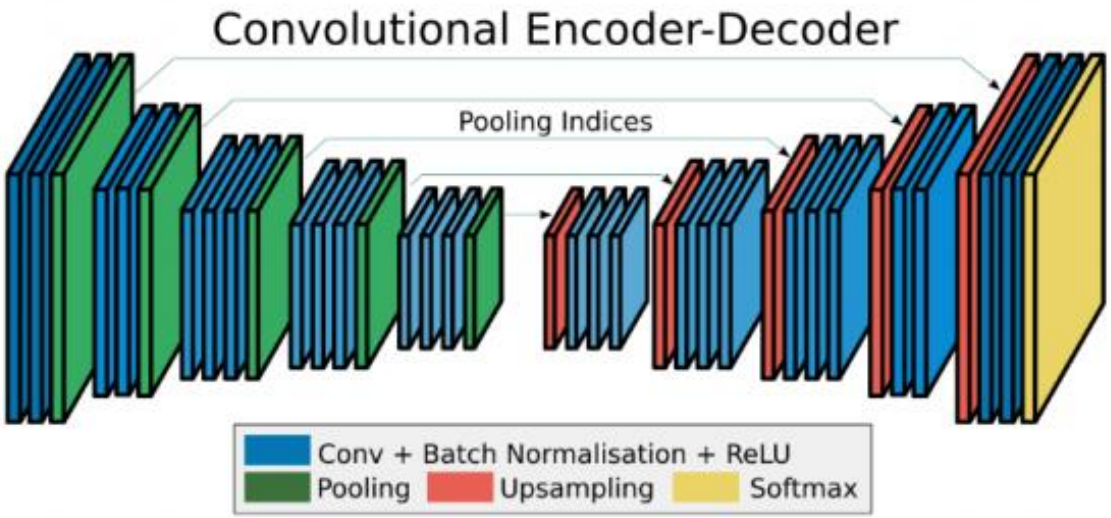
https://towardsdatascience.com/review-fcn-semantic-segmentation-eb8c9b50d2d1

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

SegNet Architecture



(Badrinarayanan, Kendall, & Cipolla, 2015)

Badrinarayanan, Vijay, Alex Kendall, and Roberto Cipolla. "Segnet: A deep convolutional encoder-decoder architecture for image segmentation." *IEEE transactions on pattern analysis and machine intelligence* 39, no. 12 (2017): 2481-2495.

In-Network upsampling: “Unpooling”

Nearest Neighbor

1	2
3	4

Input: 2 x 2

1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Output: 4 x 4

“Bed of Nails”

1	2
3	4

Input: 2 x 2

1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

Output: 4 x 4

Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

In-Network upsampling: “Max Unpooling”

Max Pooling
Remember which element was max!

Max Unpooling
Use positions from pooling layer

1	2	6	3
3	5	2	1
1	2	2	1
7	3	4	8

Input: 4 x 4

5	6
7	8

Output: 2 x 2

. . .

Rest of the network

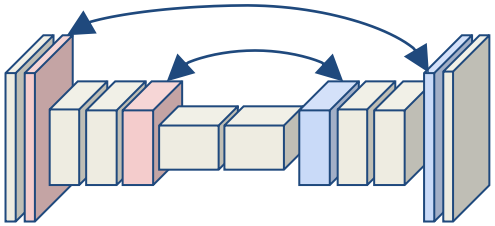
1	2
3	4

Input: 2 x 2

0	0	2	0
0	1	0	0
0	0	0	0
3	0	0	4

Output: 4 x 4

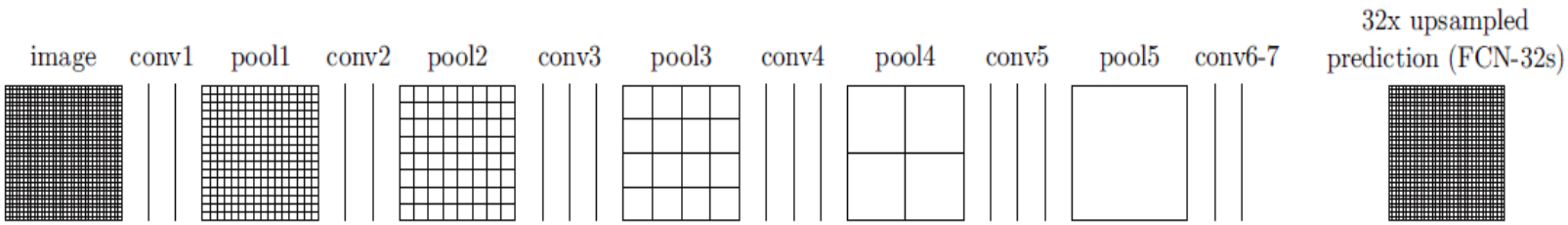
Corresponding pairs of downsampling and upsampling layers



Slide Credit: Fei-Fei Li, Justin Johnson, Serena Yeung, CS 231n

Semantic Segmentation : Fusing the Outputs

- After going through conv7 as below, the output size is small, then 32 × up-sampling (called **FCN-32s)**→ same size of input image.
  - Disadvantage : makes the output label map rough

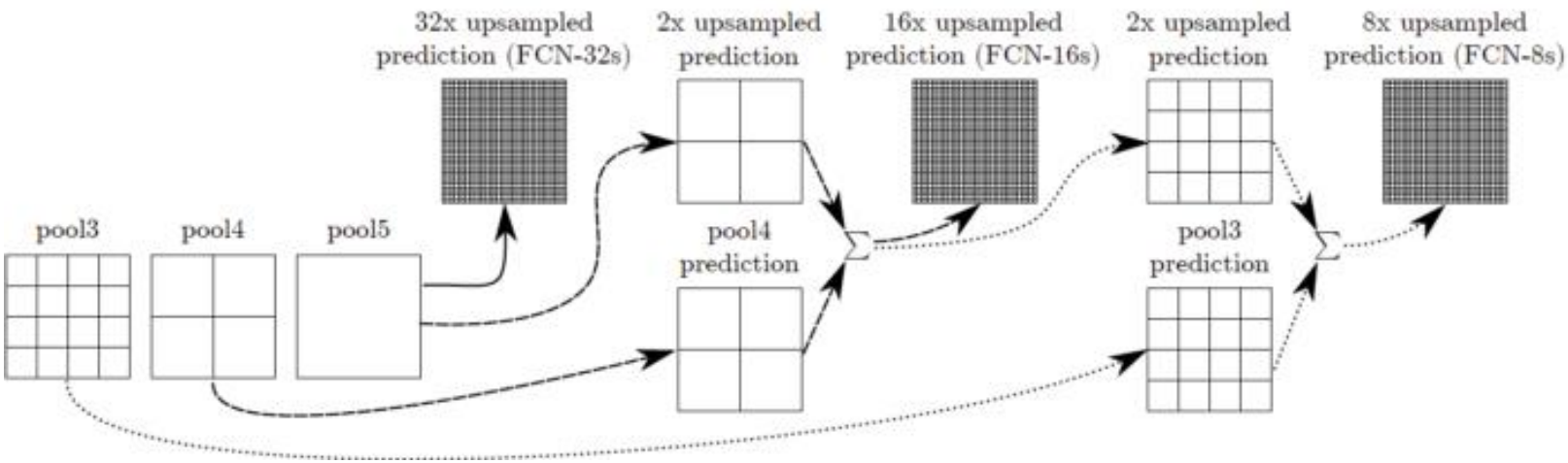


- Deep features can be obtained when going deeper, **spatial location information is also lost when going deeper**. (shallower layers have more location information).
- We combine both , we can enhance the results.

  - https://towardsdatascience.com/review-fcn-semantic-segmentation-eb8c9b50d2d1
  - https://distill.pub/2016/deconv-checkerboard/

Semantic Segmentation : Fusing the Outputs

- **FCN-16s**: The output from pool5 is 2× up-sampled and fuse with pool4 and perform 16× up-sampling. Similar operations for **FCN-8s** as in the figure above.

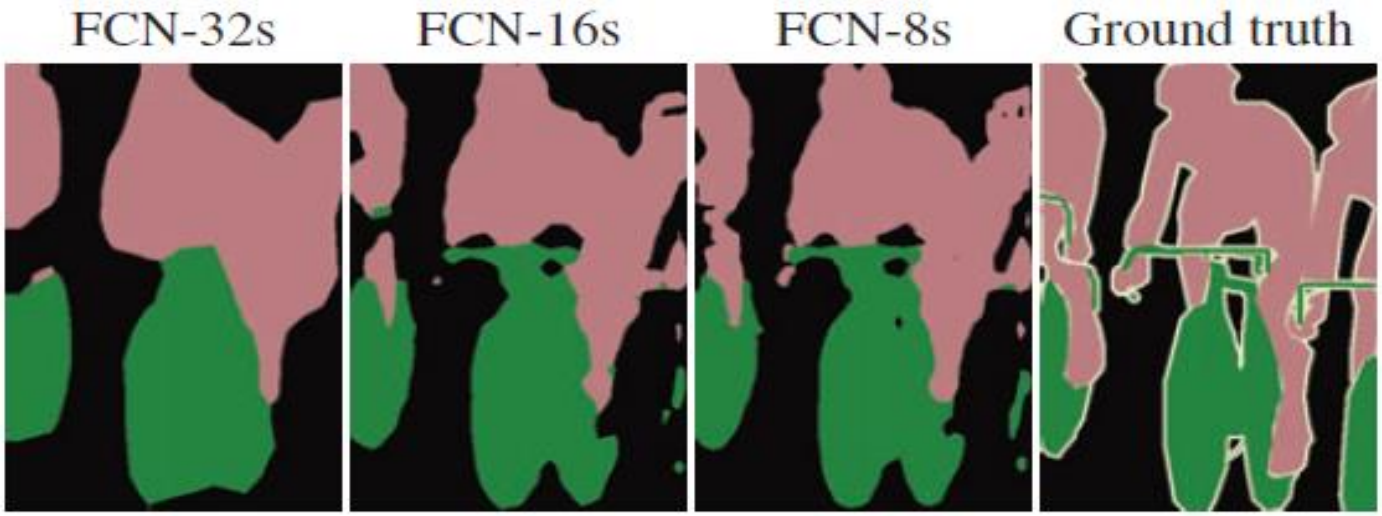


- Fusing operation actually is just like the boosting / ensemble technique used in AlexNet, VGGNet, and GoogLeNet,

- https://towardsdatascience.com/review-fcn-semantic-segmentation-eb8c9b50d2d1
- https://distill.pub/2016/deconv-checkerboard/

Comparison : Fusing the outputs

- Add the results from multiple model to make the prediction more accurate.
- Here is the comparison of fusing outputs from different layers of representation.



Comparison

U-Net Model

- One of the current state of the art models
  - Very popular published in MICCAI 2016
  - Works well with smaller number of examples (inputs and masks)
- Influenced by the previous
  - Up-conv 2x2 = bilinear up-sampling then 2x2 convolution
  - 2D slices
  - 3x3 convolutions

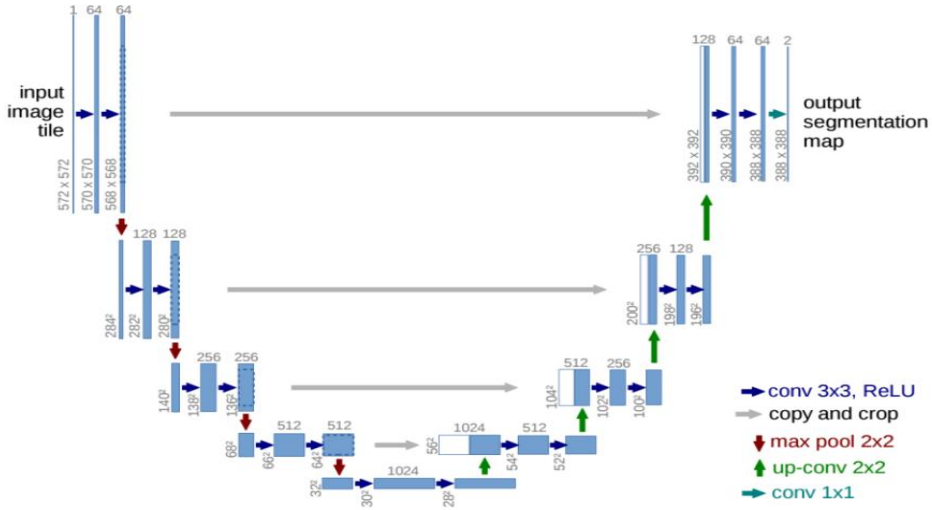


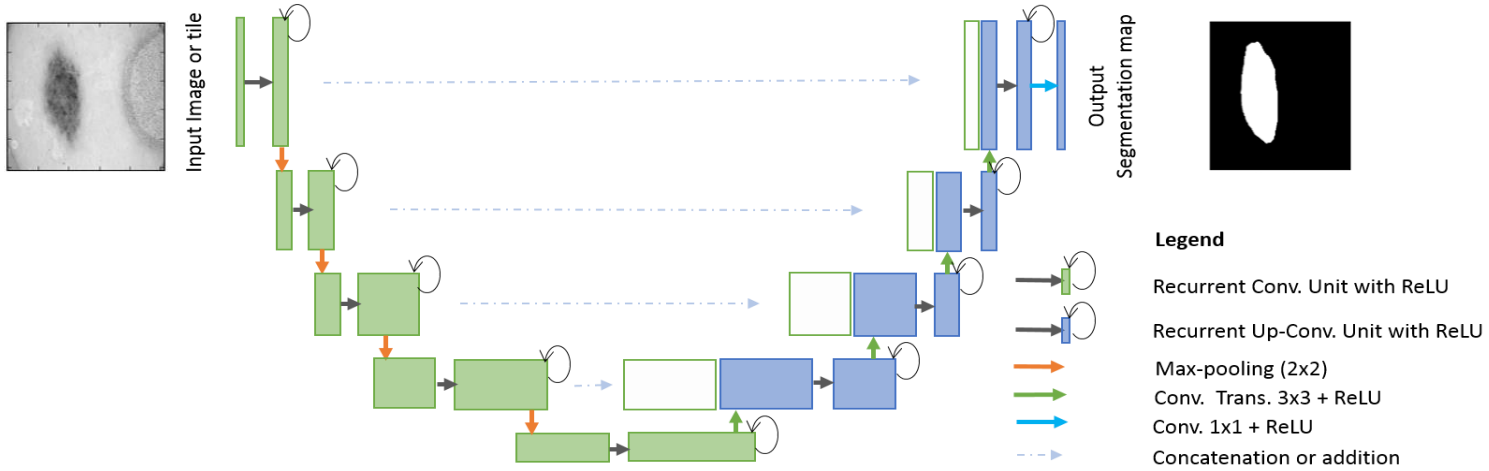
Fig. 2: U-Net: The proposed architecture by Ronneberger et al. [68].

CITED BY 134,278

Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." In *International Conference on Medical image computing and computer-assisted intervention*, pp. 234-241. Springer, Cham, 2015.

Segmentation model: R2U-Net models

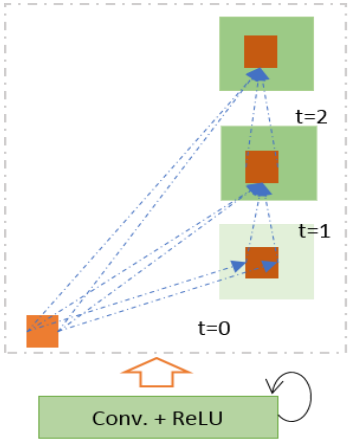
- **Advantages:**
  - Better feature representation with improved performance in segmentation tasks.
  - Efficient network with same or less number of network parameters
  - Generosity: this technique can be applied for other deep learning models such as SegNet



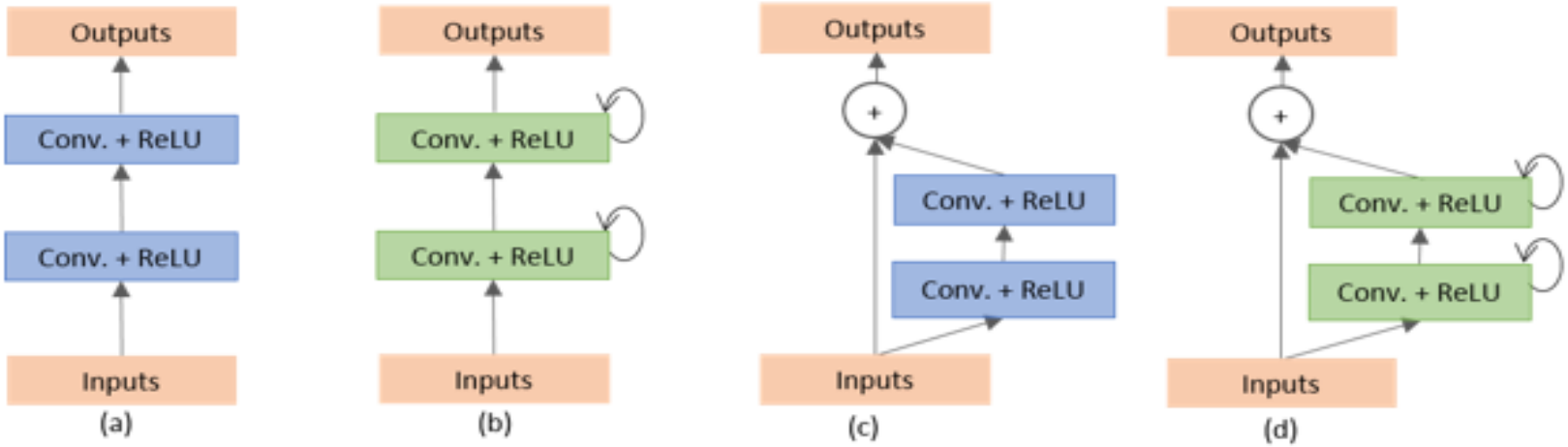
RU-Net architecture with convolutional encoding and decoding units using recurrent convolutional layers (RCL) based U-Net architecture. The residual units are used with RCL for R2U-Net architecture.

RU-Net and R2U-Net models

- U-Net architecture based on unit (a).
- Recurrent U-Net **(RecU-Net) architecture based on unit (b).**
- Residual U-Net (ResU-Net) architecture based on unit (c).
- Recurrent Residual based on U-Net models called **"R2U-Net" based on unit (d).**

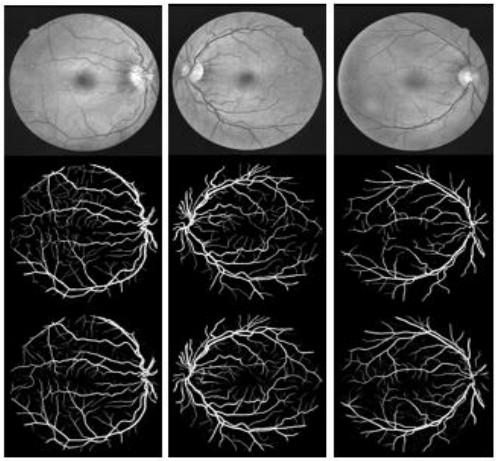


Unfolded Recurrent Convolutional layer.

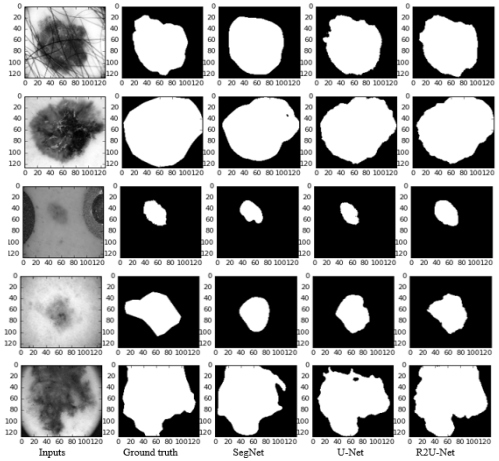


Different variant of convolutional and recurrent convolutional units (a) Forward convolutional units, (b) Recurrent convolutional block (c) Residual convolutional unit, and (d) Recurrent Residual convolutional units (RRCU).

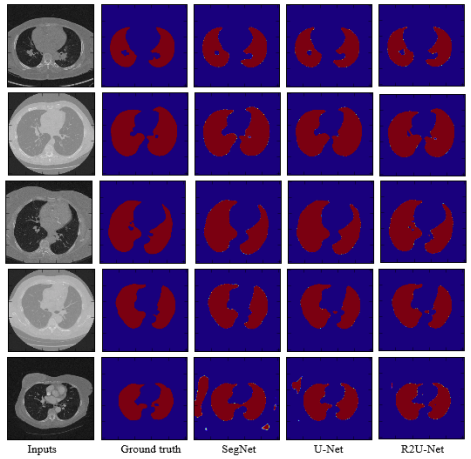
Results



Retinal Blood Vessel Segmentation



Skin Cancer Segmentation

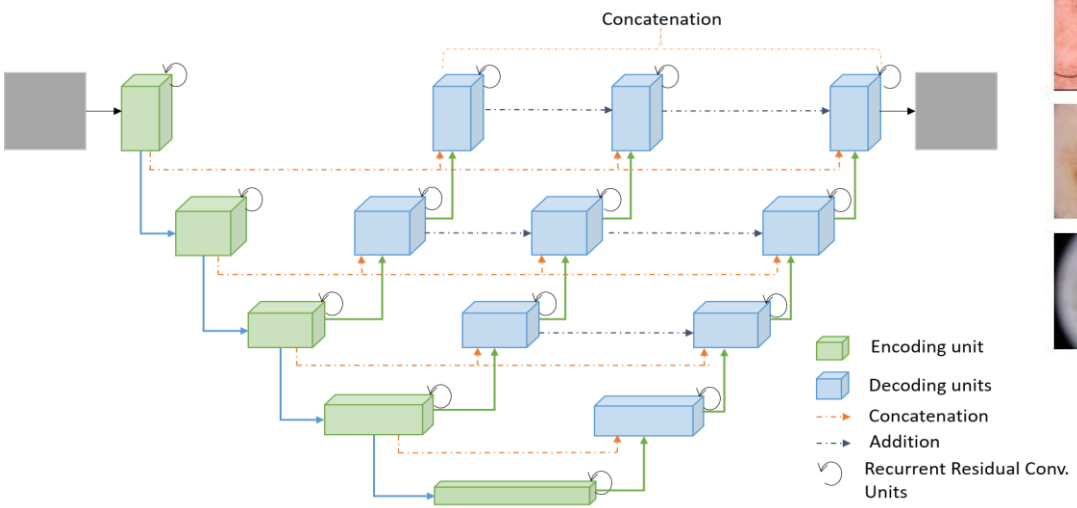


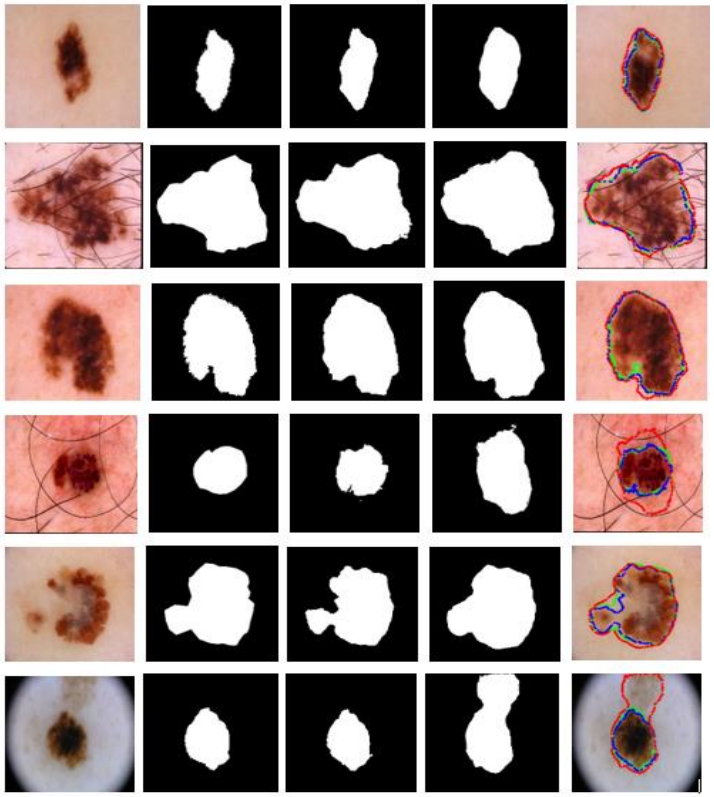
Lung Segmentation

Md Zahangir Alom, Chris Yakopcic, Mahmudul Hasan, Tarek M. Taha, Vijayan K. Asari, "Recurrent residual U-Net for medical image segmentation," *J. Med. Imag.* **6**(1), 014006 (2019), doi: 10.1117/1.JMI.6.1.014006.

NABLA-N network (∇^N-Net) model for Medical Image Segmentation

- NABLA-N network (∇^N-Net) model proposed inspired by the feature recuse concept for R2U-Net model.
- **Advantages:**
  - Efficient network with same or less number of network parameters.
  - Better performance in segmentation tasks compared to R2U-Net
  - Generosity: this technique can be applied for other U-Net based models for segmentation tasks.





Inputs GT ∇^N-Net$_{AB}$ R2U-Net Contours

Alom, Md Zahangir, Raj P. Kapur, T. J. Bowen, and Vijayan K. Asari. "GanglionNet: Objectively assess the density and distribution of ganglion cells with NABLA-N network." *Informatics in Medicine Unlocked* 23 (2021): 100518.

Dilated Convolutional Network

- The idea of Dilated Convolution is come from the **wavelet decomposition**.
- It is also called **"atrous convolution"**, "algorithme à trous" and "hole algorithm".

$$(F * k)(\mathbf{p}) = \sum_{\mathbf{s}+\mathbf{t}=\mathbf{p}} F(\mathbf{s})\, k(\mathbf{t}). \qquad (F *_l k)(\mathbf{p}) = \sum_{\mathbf{s}+l\mathbf{t}=\mathbf{p}} F(\mathbf{s})\, k(\mathbf{t}).$$

Standard Convolution (Left), Dilated Convolution (Right)

When *l*=1, it is standard convolution.

When *l*>1, it is dilated convolution.

Dilated Convolutions

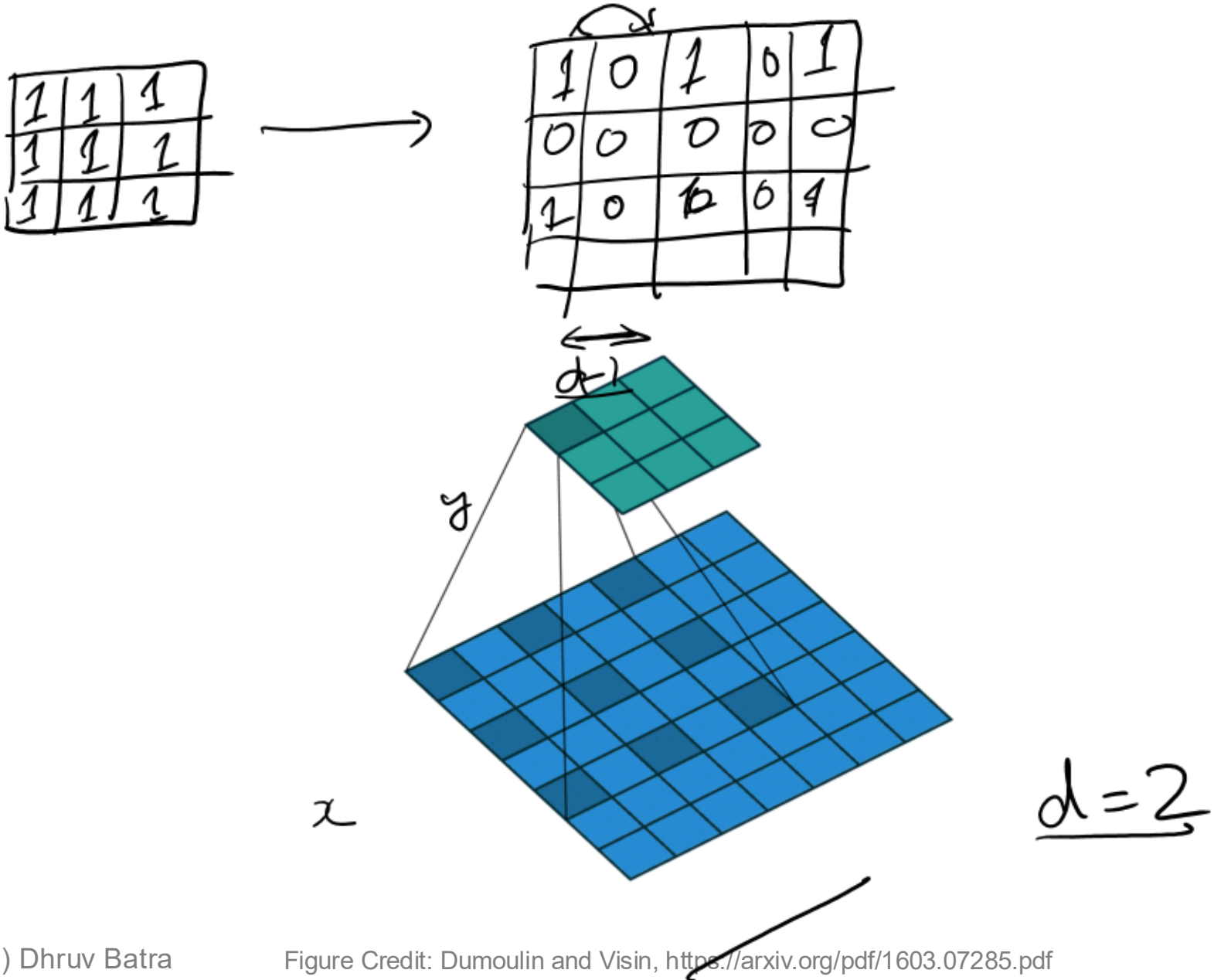
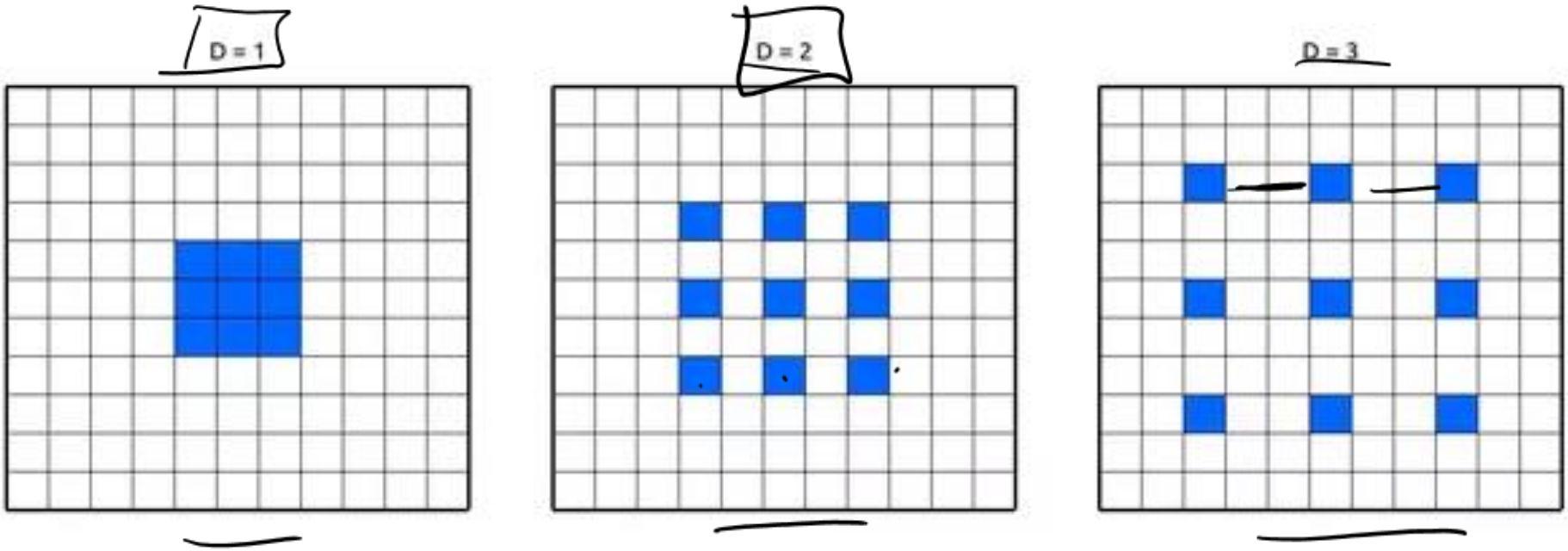


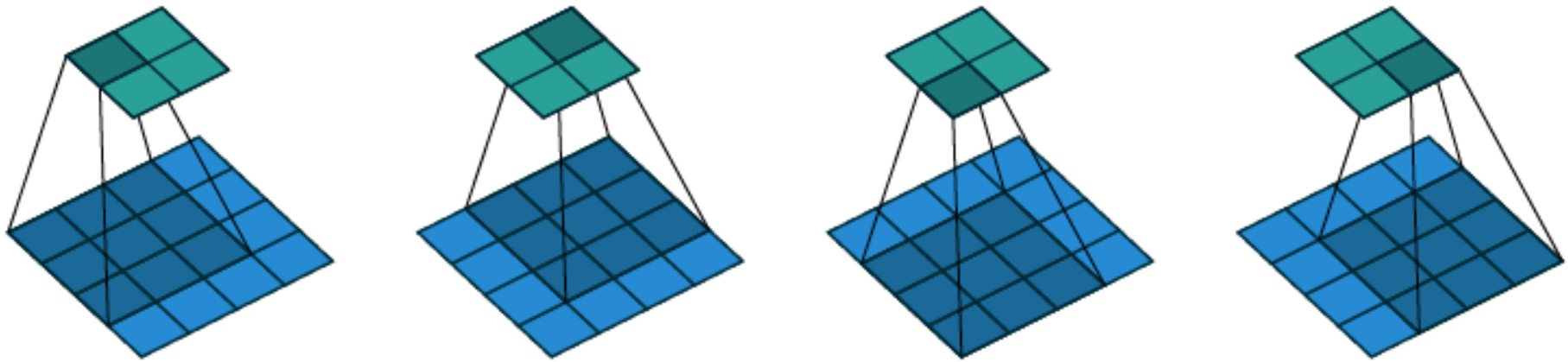
Figure Credit: Dumoulin and Visin, https://arxiv.org/pdf/1603.07285.pdf



$$k \rightarrow k + (k-1)(d-1)$$

$$3 + (3-1)(2-1) = 5$$

https://towardsdatascience.com/review-dilated-convolution-semantic-segmentation-9d5a5bd768f5

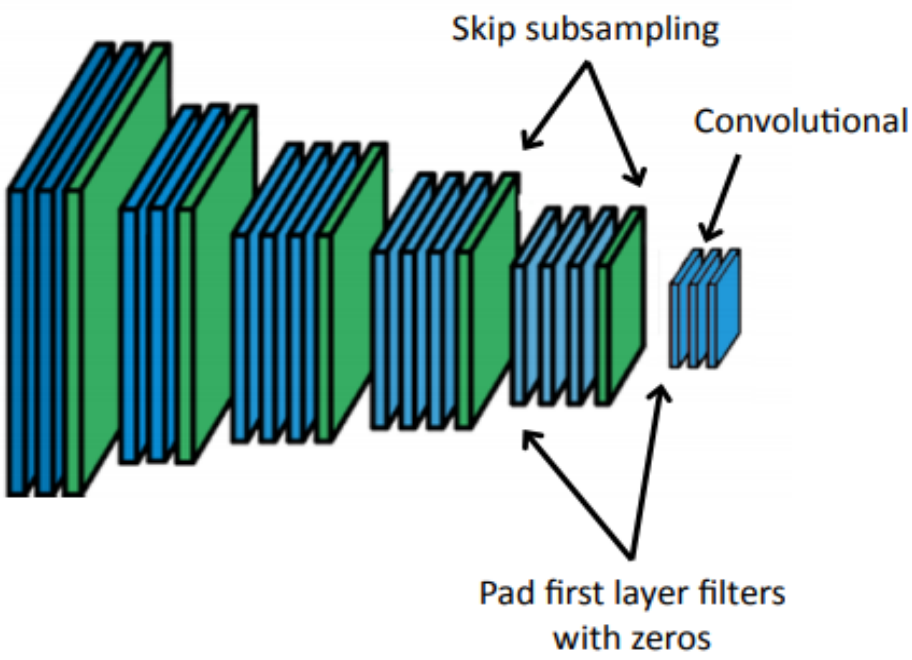


$$\begin{pmatrix} w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 & 0 \\ 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 \\ 0 & 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} \end{pmatrix}$$

Figure Credit: Dumoulin and Visin, https://arxiv.org/pdf/1603.07285.pdf

DeepLab DCNN

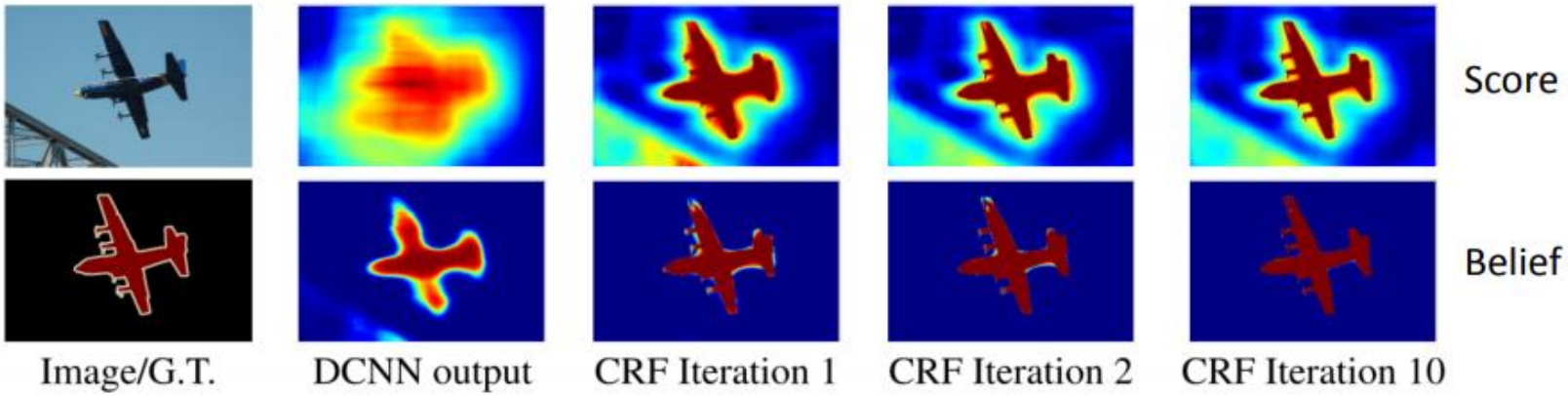
- Modify ImageNet pre-trained VGG-16 (Simonyan & Zisserman, 2014)
  - fully convolutional
  - dense features
  - Upsampling by bilinear interpolation



(image modified from Badrinarayanan, Kendall, & Cipolla, 2015)

DeepLab Conditional Random Field (CRF)

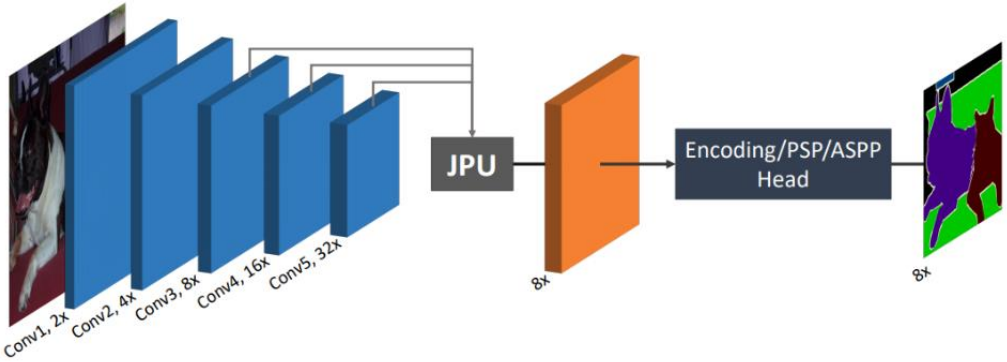
- Statistical modeling method for pattern recognition
- CRF can take context into account
- CRFs use to encode known **relationships between observations** and construct consistent interpretations



(Chen, Papandreou, Kokkinos, Murphy, & Yuille, 2014)

FastFCN: Rethinking Dilated Convolution in the Backbone for Semantic Segmentation (2019)

- Proposed a joint up-sampling module named **Joint Pyramid Up-sampling (JPU)** to **replace the dilated convolutions**
- FCN as the backbone, the **JPU uses to up-sample the low-resolution final feature maps, resulting in high-resolution feature maps.**
- Replacing the dilated convolutions with the JPU **does not result in any loss of performance.**



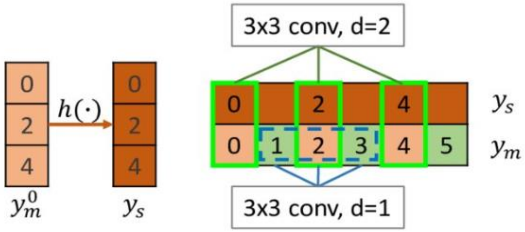


Figure 2: **Framework Overview of Our Method.** Our method employs the same backbone as the original FCN. After the backbone, a novel upsampling module named Joint Pyramid Upsampling (JPU) is proposed, which takes the last three feature maps as the inputs and generates a high-resolution feature map. A multi-scale/global context module is then employed to produce the final label map. Best viewed in color.

Wu, Huikai, Junge Zhang, Kaiqi Huang, Kongming Liang, and Yizhou Yu. "FastFCN: Rethinking dilated convolution in the backbone for semantic segmentation." *arXiv preprint arXiv:1903.11816* (2019).

JPU

- **Joint Pyramid Up-sampling (JPU)** is designed for generating a feature map that approximates the activations of the final feature map from the backbone of DilatedFCN.

- Joint up-sampling aims at generating a **high-resolution target image by transferring details and structures from the guidance image**.

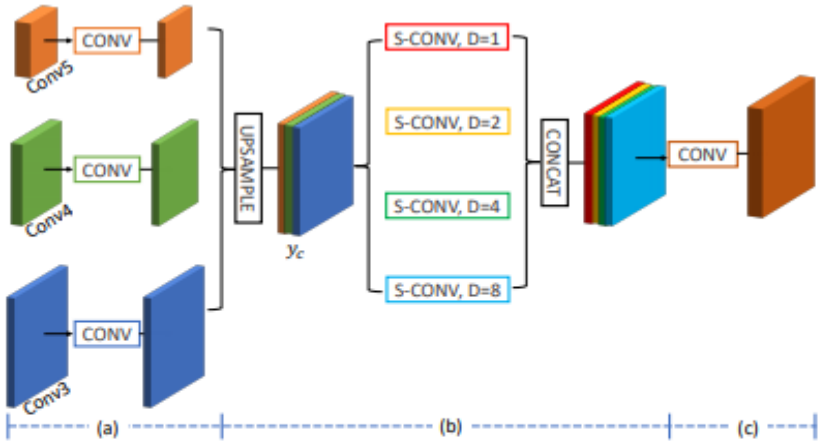


Figure 4: **The Proposed Joint Pyramid Upsampling (JPU).** Best viewed in color.

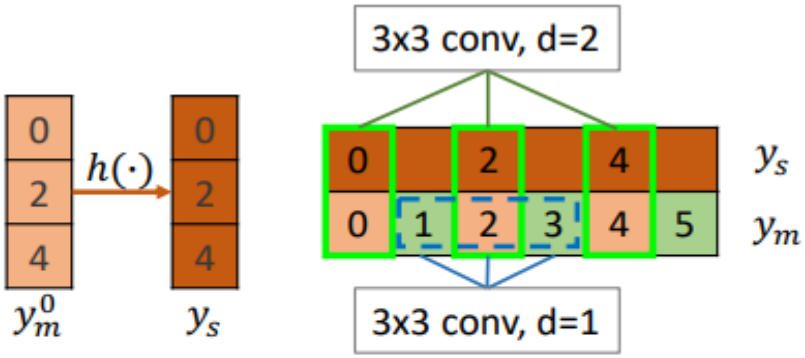


Figure 5: The convolution with dilation rate 1 focuses on y_m^0 and the rest part of y_m, and the convolution with dilation rate 2 aims at y_m^0 and y_s. Best viewed in color.

Wu, Huikai, Junge Zhang, Kaiqi Huang, Kongming Liang, and Yizhou Yu. "FastFCN: Rethinking dilated convolution in the backbone for semantic segmentation." *arXiv preprint arXiv:1903.11816* (2019).

Different between Stride and Dilated Convolution

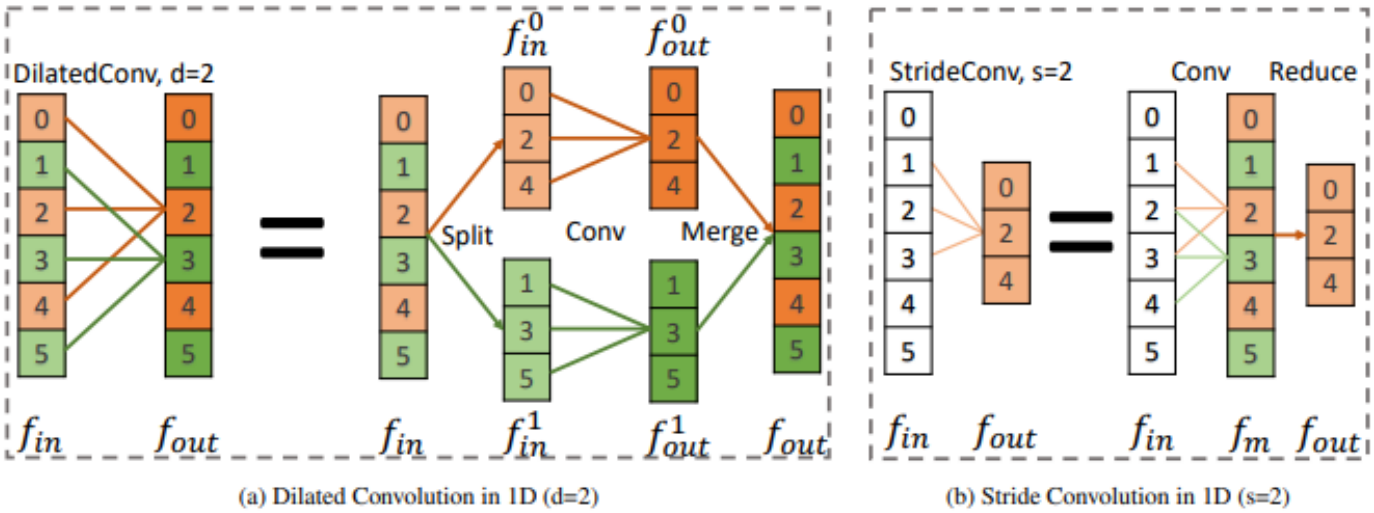
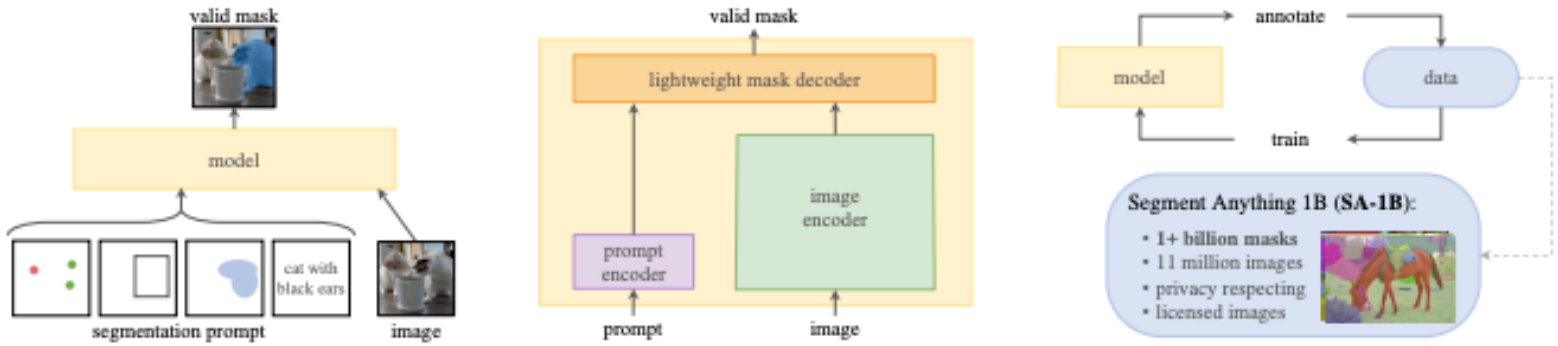


Figure 3: Dilated Convolution (dilation rate=2) and Stride Convolution (stride=2) in 1D. Best viewed in color.

- Feature Pyramid Network(FPN)
- Linknet
- Pyramid Scene Parsing Network (PSPNet)

Wu, Huikai, Junge Zhang, Kaiqi Huang, Kongming Liang, and Yizhou Yu. "FastFCN: Rethinking dilated convolution in the backbone for semantic segmentation." *arXiv preprint arXiv:1903.11816* (2019).

Segment Anything



(a) **Task**: promptable segmentation (b) **Model**: Segment Anything Model (**SAM**) (c) **Data**: data engine (top) & dataset (bottom)

Figure 1: We aim to build a foundation model for segmentation by introducing three interconnected components: a promptable segmentation *task*, a segmentation *model* (SAM) that powers data annotation and enables zero-shot transfer to a range of tasks via prompt engineering, and a *data* engine for collecting SA-1B, our dataset of over 1 billion masks.

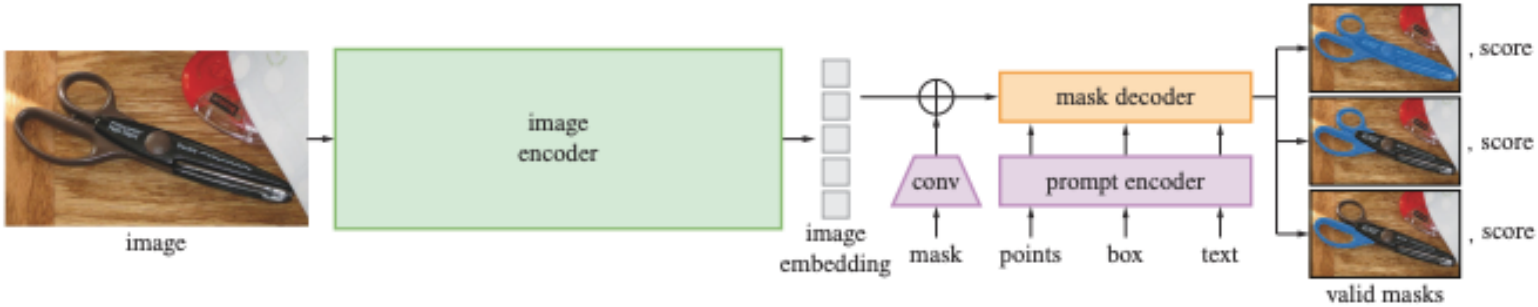


Figure 4: Segment Anything Model (SAM) overview. A heavyweight image encoder outputs an image embedding that can then be efficiently queried by a variety of input prompts to produce object masks at amortized real-time speed. For ambiguous prompts corresponding to more than one object, SAM can output multiple valid masks and associated confidence scores.

https://arxiv.org/pdf/2304.02643

Summary

- What is Regression/Segmentation models?
- Introduced Up convolution and Up-pooling
  - Deconvolution
  - Max up-pooling
- Discussed different models:
  - Fully Convolutional Networks (FCN)
  - SegNet
  - U-Net and its variants
  - DeepLab and
  - FastFCN
- Segment Anything (SAM)
- What's next?
  - Detection (classification + regression) models